

The Heart Manual

Revascularisation Edition



A six-week programme
for healthy living for
people recovering from
coronary angioplasty or
bypass surgery.

The Heart Manual: Revascularisation Edition

Foreword

NHS Lothian’s Heart Manual programme has, over the years become the leading home-based cardiac rehabilitation package in the UK and Ireland, contributing to the recovery and wellbeing of in excess 600,000 patients to date.

The programme you are about to use is similar to the original Manual in some ways and different in others. It is a special version for people recovering from coronary angioplasty or a coronary artery bypass graft (bypass). Developed in collaboration with the Heart Manual Department, NHS Lothian and commissioned by the University of Birmingham Department of Public Health and Epidemiology, it was tested with patients in a 4-year research project. Further testing by 150 patients was carried out in a hospital in South-West Scotland.

The results of this study replicated the positive results of the original Manual. We hope that this edition will bring as many benefits to those who have undergone a revascularisation procedure and their families as the original Manual has brought to those recovering from heart attacks.

As always, the Heart Manual Management Group welcomes the views of users, their supporters, families, and healthcare professionals. Please contact us with your comments.

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www.theheartmanual.com

“The UK Heart Manual (NHS Lothian) is perhaps the most extensively studied self-management book for patients recovering from acute MI or coronary revascularization.”

American Association of Cardiovascular and Pulmonary Rehabilitation,
the American Heart Association, and the American College of Cardiology, *Circulation*. 2019.

The Heart Manual (Angioplasty and Bypass/Revascularisation Edition)

11th impression

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The Heart Manual: Revascularisation Edition



This book is called the **Heart Manual: Revascularisation Edition** because it can help you understand why you needed an angioplasty, or bypass operation.

It will also tell you the things that you can do to recover in the quickest and safest way.

It can tell you about the medicines you are taking.

If you live with someone it can help them to understand your condition.

The main part of the book is divided into six weekly sections. There are some simple exercises to do, and each week there is a small section to read.

The bad news

**You know you have coronary artery disease.
You have had an operation or procedure.
You might be worried about:**

- how your condition will affect you for the rest of your life
- having a heart attack
- what you can or can't do.

**Upset emotions are common.
Most people in the same position
occasionally feel:**

- sad
- afraid
- bad-tempered
- moody.

The good news

These worries are often caused by:

- out-of-date ideas
- wrong ideas
- not knowing the true facts.

**Reading this Manual will help you to find
out the *truth* about your condition.**

**The upset feelings are *normal* and will pass
as you get better.**

Before you start the Heart Manual Programme please go to page 171 of this manual and complete your Baseline Questionnaire. You will be asked to complete a similar questionnaire at the end of the programme. This will allow you to compare how you are now, to how you are at the end of your Heart Manual Programme. This information is also very useful to us and we will ask you to return both questionnaires to the Heart Manual Department at the end of the programme.

How to use the Manual

Part 1: Your Heart Condition: the Facts

(pages 9–22)

This part is for you to read in the first days after your angioplasty, or bypass operation.



It is called:

It tells you the plain facts about your heart condition. You should read this section first, and you might want to come back to it a few times over the coming weeks.

If you live with someone, get them to read this section, too. It can help them just as much as it helps you.

Part 2: The Weekly Programme

(pages 23–138)

This part consists of six weekly sections.

The weeks are clearly labelled like this:



Each weekly section has important information to help you to recover, and an exercise programme to help you back to fitness in easy stages.

Every day:

- **follow your plan morning and evening as advised by your facilitator**
- **each time mark your activities in the charts provided**
- **practise relaxation by listening to tracks 1 and 2 on the relaxation CD or audio.**

You can access the audio online here:

www.heartmanual.scot.nhs.uk/index.php/download/resources

or scan this QR code



or download the HM Relax app via Apple or Google Play stores or scan the QR code. Please ask your facilitator for the username. The password is relax.



Part 3: Facts and Advice to Help Your Recovery

(pages 139–179)

This part provides extra information about your recovery, medicines, wound healing, and other things you might want to know. You can read it right through or dip into it to answer your questions.

The Heart Manual: Revascularisation Edition

Not just a book

This Manual isn't just a textbook of information. It is a proven rehabilitation tool – an important part of your treatment which has been demonstrated to help people in the first weeks after their angioplasty or bypass operation.

It is designed for you to use as you wish and for **you** to adapt to **your** needs, with regular support.

This Manual has been given to you by a healthcare worker (usually a **nurse**) who has been specially trained to work with patients using the Manual. We call this person a **facilitator**.

In the first 6 weeks at home, you will be contacted by another facilitator. He or she can discuss with you how you are getting on with the Manual, answer any questions you may have, and discuss future contact dates and times.

You might like to write down the names of your facilitators here:

Hospital Facilitator

Name _____

Contact phone number _____

Community (home) Facilitator

Name _____

Contact phone number _____



Important message

If you know someone else who has a heart condition, **please do not show them this Manual or the CD/audio.**

Do not advise them to do any of the exercises or give them advice from the Manual. You may be very tempted to help them, **but please don't. Instead of helping them you may harm them.**

This is very important because they may have a different or much more serious kind of heart condition. **Your advice may be wrong for them and you may cause them harm.**

There is no way that you can guess about this. This Manual is only suitable for people who have been advised to use it by trained healthcare staff.

None of the advice in this Manual is intended to replace any advice that you are given by your doctor.

Part 1

Your Heart Condition: the Facts

Have you completed your baseline questionnaire on page 171? If not please take a few minutes to complete it now. You will be asked to complete a similar questionnaire at the end of the programme. This will allow you to compare how you are now, to how you are at the end of your Heart Manual Programme.

Starting your recovery

What this Manual is for?

This Manual is designed to help you to make important changes to your lifestyle which can improve your coronary artery disease. By changing the way you live, you can:

- feel much better
- reduce the chance of having further heart problems
- improve the condition of your arteries
- help to prevent other health problems

The Heart Manual was originally designed for people who have had heart attacks. For many of them it has helped to transform their lives. Many of them are now fitter, happier and healthier than they were before their heart attack, and this makes them better able to fight off a lot of other diseases as well.

Who is this Manual for?

The people who use this Manual all have the same heart condition – coronary artery disease.

You will learn much more about this disease in the following pages and in the weekly programme in Part 2.

Although everyone who is given this Manual has the same disease, they haven't all had the same treatment.

This Manual has been designed to help those who have had two different procedures:

- Those who have had percutaneous transluminal coronary angioplasty or PTCA. This procedure might also be referred to as percutaneous coronary intervention or PCI. In many cases this procedure also includes the insertion of a stent OR
- Those who have recently had a coronary artery bypass graft operation. This operation is sometimes shortened to CABG – you may hear it pronounced 'cabbage' for short.

To make things simple we shall either refer to angioplasty or coronary artery bypass graft (bypass) individually or refer to them both as 'your procedure'.

You and your heart condition

The first section of this Manual gives you the plain straightforward facts about your heart condition and the treatment you have had for it. It answers some of the questions that people usually ask. We suggest that you read through this section in the early days after your procedure. If you live with someone, they should read it, too.

As soon as you can – we suggest straight away after angioplasty or as soon as you get home after a bypass – move on to Part 2 (page 23) and start the weekly programme.

But please remember not to show this Manual to anyone else who has a heart condition. This is because:

- **their heart condition may have a different cause**
- **they may have a much more serious problem than yours**
- **the advice may be wrong for them**

Only a healthcare professional can decide this.

What was the procedure that I had?

Your procedure

Before you had your procedure, your cardiologist or surgeon will have explained in some detail what was going to happen. People are often rather anxious at this stage, and sometimes find all the information too much to take in. Doctors also sometimes use complicated and difficult language which some people find hard to understand. Don't be afraid to ask them to explain again if you are unsure what they mean.

Why did I have one procedure and not the other?

The main reason why people have either procedure is to improve the circulation of blood through the arteries in their heart. Both procedures have been proven over many years to work well.

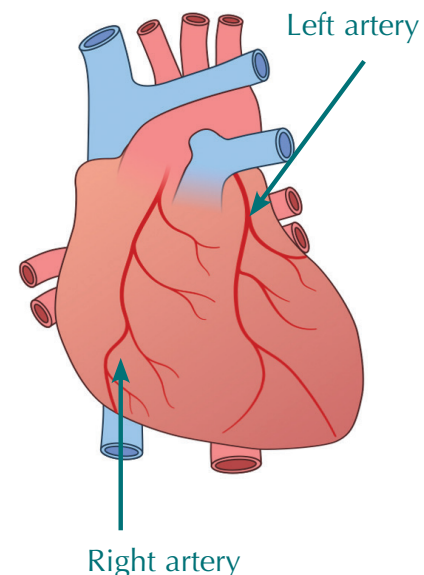
Angioplasty (drug eluting stent or balloon) is done by the cardiologist, most often as a follow-on at the same time as an angiogram (X-ray) investigation. If angioplasty was not suitable for you, you may have been referred to a surgeon for a **bypass** operation.

The reason you have had one procedure and not the other depends on many factors, such as:

- **how many narrowings there were in your arteries**
- **where the narrowings were and how severe they were**
- **your previous medical history**
- **which procedure was best for you.**

If the person in the next bed or your neighbour or a relative has not had the same procedure as you, this doesn't necessarily mean that your condition is better or worse than theirs. It's just a different treatment. Some people have both. Try not to compare yourself with other people.

What's wrong with my heart?



Most people who need an angioplasty (stent) or bypass have nothing wrong with their heart. When they are told they need a procedure, many people believe this means that their heart is 'worn out' or is about to 'pack up.'

This is completely untrue. It doesn't mean your heart is worn out or that you are 'finished.'

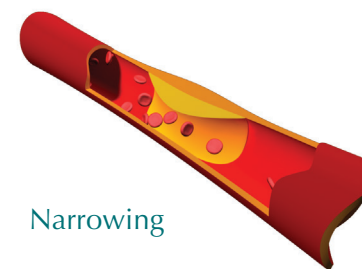
Your heart is a muscle which pumps blood around your body. Like all muscles, it has its own blood supply through arteries which run over its surface. These are called coronary arteries.

If any of the arteries become narrow, the blood flow to the heart is reduced. This can cause symptoms such as angina. Angina is a type of cramp in the heart muscle. Some people experience angina as a tightness or discomfort in the chest rather than a pain. Occasionally discomfort may be felt in the arm, the back or the jaw. While others may experience shortness of breath. All that the procedure does is to get

the blood flowing properly again - either by putting in a new blood vessel for the blood to flow through (bypass graft), or by widening the arteries that have become narrow (stent or balloon). The angioplasty or bypass does not do anything directly to the heart. The heart is usually working fine. It's the arteries that are the problem.

The other important thing about narrowed arteries is that they can get blocked completely. If this happens suddenly, it can cause a myocardial infarction (MI) – what most people call a heart attack.

What caused the narrowing?



You have **coronary artery disease**. This means a gradual thickening of the walls of the arteries, which makes the insides become narrower. This happens over a number of years. Many people have coronary artery disease without knowing it. Often it doesn't cause any harm.

The thickening is mostly caused by fatty deposits made of cholesterol which build up in the walls of the arteries. It can get worse if a narrowed part of your artery gets damaged or inflamed. When it heals it can leave a scar, and this scar tissue can make the artery even narrower.

Sometimes an area of artery becomes so narrow that it can't carry as much blood as your heart needs.

You will have noticed that the whole story so far is about your **arteries** and not about your **heart**. All your heart has done so far is to let you know by giving you angina that it isn't getting enough blood. So angina is caused by coronary artery disease – not because of a problem with the heart.

Were all of my arteries narrowed?

No. Coronary artery disease usually affects parts of one or a few arteries, but it rarely extends along the whole length of an artery. It may just have been one or two narrow areas that were causing the problems.

What's been done to me?

This Manual isn't about your procedure - it's about your recovery. Before your angioplasty or bypass your cardiologist or surgeon should have discussed in detail with you and your partner or carer what was going to be done to you, and you were probably given a patient information booklet describing the procedure. Just in case you've lost the booklet or forgotten what the doctor said, you will find a quick description of what happened on the following pages:

For **Angioplasty** see page 15-16

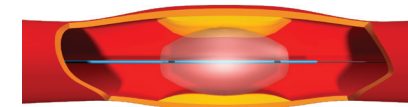
For **Coronary Artery Bypass Graft (bypass)** see page 17-19

For general tips which apply no matter what procedure you have had, see page 20

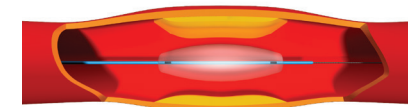
Your Procedure: Angioplasty



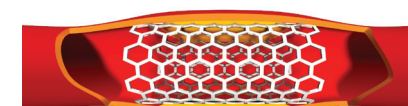
Catheter



Balloon inflated



Balloon deflated



Stent

As you know, your coronary artery disease meant that cholesterol and scar tissue had built up on the walls of your coronary arteries, making them narrow. The purpose of an angioplasty procedure is to stretch the artery wall at the narrowed points, so there is more space for blood to flow through.

A special type of catheter with a small balloon on the end was threaded through an artery in your arm or at the top of your leg and passed up into your coronary arteries. Using an X-ray camera, the balloon was positioned in the narrow part of your artery. It was then inflated with enough pressure to stretch the artery where the coronary disease had narrowed it. The balloon was then deflated, and dye was injected to check if the narrowing had been stretched wider. The balloon may have needed to be inflated several times to get the best result.

Your cardiologist may have used a drug-eluting balloon (DEB), which is coated with medication that is released into the artery wall during inflation. This helps prevent the artery from narrowing again and does not leave anything behind. DEBs can be useful if a previous stent has narrowed again or if placing a new stent is not the best option.

The cardiologist may also place a small metal tube called a stent inside the artery to help keep it open. The stent is stretched by the balloon to fit the exact size and shape of the inside of the artery. Once expanded, it works like scaffolding to hold the artery open and eventually becomes part of the artery wall, reducing the chance of future narrowing.

Starting your recovery

Your angioplasty should relieve your angina very quickly and you might think:

'I'm feeling much better – I don't know what all the fuss was about!'

You might get back to your normal life within a few days and feel like carrying on as if nothing had happened. You might think:

'There's nothing wrong with me any more so I don't need to bother with this Manual.'

If you do think this, you're mistaken. Find out why in the next section.

Angioplasty can make you think you're cured

But you should still follow the Manual

We hope that you will decide to carry on with the Manual. But some people who are used to leading active lives might find parts of the Exercise/Activity Plan easy.

That's fine. You might decide as you work through the weekly programme to set yourself a harder target than what is shown in the Exercise/Activity Plan. How fit or unfit you were before the procedure is not the important thing.

The Manual helps you to fight back

What is important is the fact that you have coronary artery disease. Following the Exercise/Activity Plan and the other advice in the Manual can help you to get fitter and then to stay fit.

No matter what point you start from, the more you take regular exercise, and the more you do the other things recommended in the Manual, the more you will reduce the chances of your symptoms returning or needing further treatment in the future.

How you feel afterwards

How you feel afterwards depends on a number of things:

- **How your angioplasty procedure went.** If there were no medical complications during or after your procedure you will probably feel better quite quickly. If however, your procedure took more time or was more difficult than expected you may need a little more time to recover. It is normal to experience some localised chest discomfort/ache in the early days, but this should ease quickly.
- **How fit you were before the procedure.** If your everyday life was badly restricted by coronary artery disease before your procedure, it may take longer for you to gain the full benefit of the new blood flow to your heart.
- **How confident you are of making a full recovery.** This Manual will help you with this, using a steady paced approach to feel better mentally and physically.

This Manual has been designed to help you recover and prevent further problems in the future.

What to do

- Check your wound is healing well. Follow the wound care instructions given to you by the hospital. It is important to inform your doctor or nurse if:
 - Your wound becomes increasingly painful, swollen and red, or starts to ooze or bleed.
 - You develop a temperature.

Some common concerns

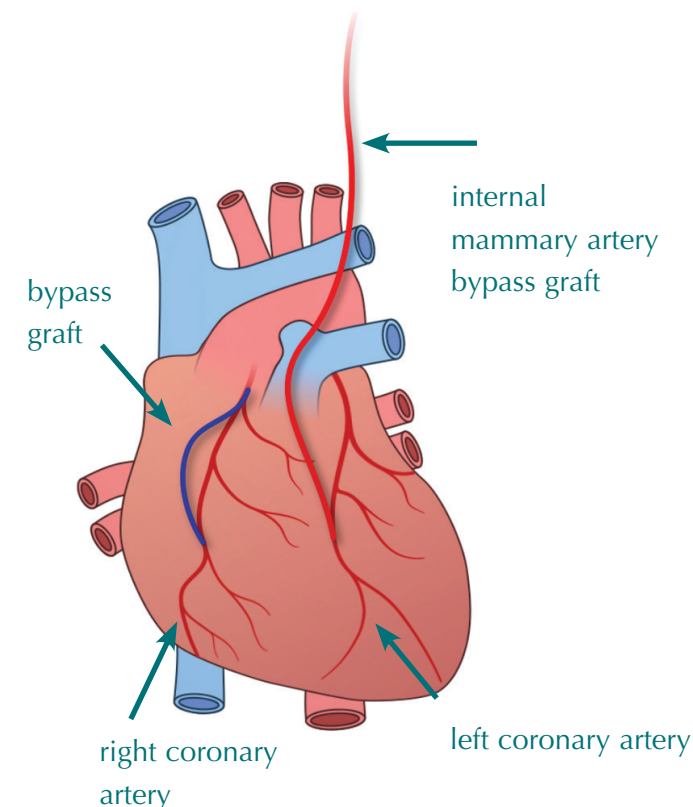
- Worry (anxiety) and sadness (depression) are common. For practical advice on dealing with these problems, read the relevant sections in Part 3 of the Manual (pages 156–169).
- You may find that your breathing feels different, particularly if you were short of breath due to the narrowing in your artery. It may also be because you have felt anxious or worried. Read the sections on Breathlessness and Hyperventilation in Part 3 of the Manual (pages 154–155).

Starting your recovery

Your Procedure: Coronary Artery Bypass Graft (CABG)

Bypass surgery is a major operation but a very common one, with thousands being performed in the UK each year. It has a very high success rate.

The idea is to change your heart's plumbing (blood supply) by using a different blood vessel to re-route the blood around the narrowed artery. This will have been done in one of two ways (or sometimes the two techniques are combined in one operation).



- Running down either side of your breastbone are two arteries called the **internal mammary arteries (IMAs)**. These normally help to supply blood to your chest wall. They run very near to the coronary arteries, and the surgeon can free the end of the IMA and stitch it on to the coronary artery downstream from the narrowing. You then have a new blood supply!
- The second option involves removing a blood vessel from your leg or arm which is used to bypass the narrowed section. The surgeon stitches one end to the main artery leading from your heart just next to where your own coronary arteries start. The other end is then stitched to the coronary artery downstream from the narrowing.

During your bypass operation you would have been given a general anaesthetic. Your chest would have been opened up either through the breastbone or just under your ribs, and your blood circulation may have been handled by a heart–lung bypass machine.

A CABG operation usually requires about a week in hospital.

Starting your recovery

In time your bypass operation should make you feel much better, but first you have to recover from the effects of a major operation. For some weeks or even months to come, you might be thinking to yourself:

'Is it my operation that is making me feel this way or is there still something wrong with my heart?'

You need to recover from your operation

The operation has a very high success rate

If you find yourself thinking this, remember that the procedure has a very high success rate. It's almost always the after-effects of the operation that cause the pain or discomfort.

No wonder you don't feel well!

Think about what was done in the procedure. You were given an anaesthetic. The surgeon opened up your chest to get to your heart. Then your chest may have been wired and sewn back together again. You might also have had cuts made on your leg or arm to remove veins or arteries. You were flat on your back in an uncomfortable position for several hours. It would be very surprising if it didn't take you several weeks or months to recover.

How can I tell what is because of the operation and what is because of my heart?

It is often difficult for people to believe and then to remember that most aches and pains after a bypass operation are caused by the operation itself. This means that you might feel anxious and sad from time to time.

Anxiety and sadness are very common among people with heart conditions. These feelings can be caused by wrong ideas and beliefs about 'heart trouble.' You can do a lot to fight back against them if you know the true facts.

Starting your recovery

How you feel afterwards

How you feel afterwards depends on a number of things:

- **How your operation went.** If there were no medical complications during or after your procedure you will probably feel better quite quickly. If however, your procedure took more time or was more difficult than expected you may need a little more time to recover. But remember recovering from this procedure often takes **at least** two to three months.
- **How fit you were before the operation.** If your everyday life was badly restricted by coronary artery disease before your operation, it may take longer for you to gain the full benefit of the new blood flow to your heart.
- **How confident you are of making a full recovery.** This Manual will help you with this, using a steady paced approach to feel better mentally and physically.

This Manual has been designed to help you recover and prevent further problems in the future.

What to do

- Let your family doctor know that you are home from hospital – don't just assume that the hospital will tell the doctor for you.
- After a bypass, it may help to have someone stay with you for the first week or two when you get home. If you live alone and need some extra help, you should discuss this with the hospital staff before you are discharged. Your GP may also be able to advise you.

Starting your recovery

- Check your wound is healing well. Follow the wound care instructions given to you by the hospital. It is important to inform your doctor or nurse if:

Your wound becomes increasingly painful, swollen and red, or starts to ooze or bleed.
You develop a temperature.

Some common concerns

- Don't expect your breast bone to be completely better for about 3 months (see Part 3, page 140).
- Don't be surprised if you feel very tired. This is because recovering from a major operation is a tiring business.
- Expect to get quite a bit of muscle and joint discomfort from the centre of your chest, neck, back and arms – the areas that were put under most stress during the operation. Gentle exercise can help to loosen you up and settle the pain and stiffness more quickly as will the relaxation exercises (see Part 2, pages 35-37).
- If a blood vessel was taken from your leg or arm for your bypass graft, you might have some pain or discomfort, or numbness or pins and needles there. This will pass in time. If you were given an elastic support stocking you should wear it as advised. It may also help if you keep your feet raised when sitting and try to avoid crossing your legs (see Part 3, pages 141-142).
- You will have been given a number of medications when you left hospital. Some may be for controlling discomfort or pain. Remember to take these regularly following the instructions given. If you feel that your pain is slowing down your recovery speak to your doctor.
- You may find that your breathing feels different. If you were breathless before your operation you may need to re-learn to breathe normally after the operation. Even if you weren't breathless before you may feel that way now because people tend to take shallow breaths after the operation to avoid pain. Read pages 154-155 in Part 3 of the Manual, on Breathlessness and Hyperventilation and listen to the breathing exercises on the relaxation CD/audio.
- Some people report that they have a bad taste in the mouth. This can be a short-term side-effect of the operation, but may also be due to taking different kinds of medicines.
- Sleeping difficulties are quite common after a bypass operation. This can be due to sleeping in a different position because of your chest wound. Or it can be because you may be less active for a few weeks after the operation, and you simply aren't physically tired enough to get a good night's sleep (see Part 3 page 170).
- Worry (anxiety) and sadness (depression) are common. For practical advice on dealing with these problems, read pages 156-169 in Part 3 of the Manual.
- Some people find that they have some loss of memory or concentration after a bypass operation. This is usually only temporary, and you can expect it to improve over the next few months.
- Some people notice slight changes in their eyesight. This may last a few weeks, so avoid having your eyes tested for about 3 months after your operation unless your sight suddenly changes or is becoming worse.

If you think you aren't recovering from your operation as quickly as you expected, it is important to remember that different people recover at different speeds. Some people recover in about 3 months, but it often takes longer. The speed of your recovery is influenced by a range of different factors.

Has the procedure cured the disease?

Remember that your procedure improved the blood supply to your heart – it didn't cure the disease.

You will probably feel much better either straight away or quite soon. The angioplasty (stent) or bypass will have worked almost immediately bringing extra blood flow and more oxygen to your heart. In most cases it gets rid of angina there and then.

Sometimes the heart has to adjust to the changes in blood supply and 'divert' the extra flow of blood to the parts of the heart that need it most. This might mean that it takes a little longer for your symptoms to disappear completely.

Many people soon feel that they can do more without bringing on symptoms, but if you have had a bypass it may take several weeks to recover from the effects of the surgery. However, most people feel able to do more than they have done in years.

It is important to look after your arteries so that they don't become blocked or narrowed in the future. That's why now is a good time to make some real changes in your life.

Will I get worse again or have a heart attack?

There are a lot of wrong ideas about angioplasty and bypass

Some people think: 'Angioplasty or bypass procedures only last for a few months (or years) and then the angina always comes back.' Wrong!!!

Others think: 'Whatever the doctors say or do, once you have angina all you can look forward to is more pain followed by a heart attack followed by death.'

Wrong again!!!

Some optimists think: 'Having your arteries fixed guarantees you for the rest of your life whatever you do.' And wrong again!!!

There are no guarantees for anyone

There are no guarantees for anyone – for better or for worse.

But the majority of people who help themselves, by changing some of the things that they do, get rid of angina for good.

You will find out much more about **coronary artery disease** – and what you can do about it – in Part 2 of this Manual.

Things to remember

- The problem is in your arteries and not in your heart.
- Your heart is the strongest muscle in your body.
- No other muscle has the ability to work non-stop 24 hours a day.
- Because it works all the time it is a very active muscle.
- Because it works all the time it is capable of doing much more work than most of us ever need.

Some general tips following Angioplasty and Bypass

- **Driving:** The advice given about returning to driving varies depending on your diagnosis, procedure and the type of driving licence you hold. Some drivers may even require additional tests before they can drive again. All drivers should let their insurance company know about their condition to ensure they are covered if they need to make a claim. Before returning to driving it is best to get some advice from your doctor or discuss it with your facilitator. Legal guidelines change from time to time so it is important you get the correct information. You can also check the latest information on driving and medical conditions at www.gov.uk
- **Going back to work:** For advice on going back to work if you are employed, read page 135 in Part 2 of the Manual.
- **Sex:** For advice on sex, read pages 123–124 in Part 2 of the Manual.
- **Flying:** There's no reason why you shouldn't fly once you have made a full recovery. Recent guidelines suggest that the majority of patients with heart conditions can travel safely, although some may require additional assistance. It is important to check with your GP and facilitator before planning any trips and you are advised to inform your insurance company about your condition and the procedure you have had.

For further information on fitness to fly you can consult the Civil Aviation Authority website www.caa.co.uk

Remember to make sure that you have the right amount of medication for the duration of your trip, and always carry it in your hand baggage. Make sure you take advice if you are going to miss any hospital appointments.

Why ... ? How ... ? When ... ?

You probably have many more questions to ask. It may help to write them down here and ask them the next time you speak to your doctor or facilitator. Write down the answers, too. Don't be afraid to ask what you think is a 'silly question.' It is important to get questions cleared up so that you don't dwell on them. Your doctor or facilitator will have heard it before!

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Part 2

The Weekly Programme

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Week 1

Getting Home - Getting Better



The Heart Manual

Revascularisation
Edition

Week 1:

Getting Home- Getting Better

Welcome to the first week of your 6-week recovery programme

- In this first week we shall:
- answer some more questions about your condition and why you feel the way you do
 - introduce you to the Exercise and Relaxation Plans which will play an important part in your recovery and afterwards
 - look at some of the things which might be worrying you, and show you how you can fight back
 - if you live with someone, get them to read this section - it can help them as much as it helps you
 - if you haven't done so yet fill in your baseline questionnaire on page 171.

I still don't feel well

After an angioplasty

It is normal to experience some short-term chest discomfort in the early days, but you should feel better soon.

Any bruising in the place where the catheter was put in will gradually settle, but if you notice any increase in pain, redness or swelling, tell your doctor.

After a bypass operation

Even though your heart has a new blood supply, it will take a few weeks for you to feel more energetic. An operation can often make you feel very tired afterwards because your body is using so much of its energy to heal your wounds. In these early stages be sure to pace your activity to avoid getting over-tired. Try to take a nap after activities like eating or going for a walk.

It's important not to be too worried if you feel tired or breathless – this will gradually improve with time. However, if it becomes worse rather than improving, you will need to contact your doctor.

After a major operation it is quite natural to notice a lot more sensations coming from the chest or stomach. This may be because you are paying more attention to that area. Some of these feelings would have passed unnoticed before your operation.

It's normal to feel pain or discomfort from the wound in your chest. If a blood vessel was taken from your arm or leg it may feel sore and puffy for a while. Gentle exercise and elevating the area while sitting will help to reduce the swelling. If you have been given a support stocking remember to wear it as recommended.

I'm feeling upset and unsettled

This is understandable. You have just had a procedure and it is normal and natural to feel this way.

Should I take things easy?

While you are recovering it is sensible not to overdo things.

Have a look at the 'Do I or don't I?' section on the next page.

But remember:

- **the problem is in your arteries and not in your heart**
- **your heart is the strongest muscle in your body**
- **no other muscle has the ability to work non-stop 24 hours a day**
- **because it works all the time it is a very active muscle**
- **it is also capable of doing much more work than most of us ever need.**

Think of the world's best athletes. Your heart is not much different from theirs. It has plenty of spare capacity.

So you don't have to treat yourself like an invalid.

Why do I still need all of these medicines?

Different medicines do different things.

Some medicines help to reduce the **risk factors** which cause coronary artery disease. You'll be hearing a lot more about these in the following weeks of the programme.

Some medicines reduce the risk of having a heart attack. For instance, **aspirin** makes your blood slightly less 'sticky', so if you did form a clot in the future, it is less likely to become large and block the artery.

Other medicines help your heart to recover better. Pages 142–149 in Part 3 are full of useful information about the medicines and what they will do to help.

If you have had a bypass operation, you may be on medicines to control your heart rhythm, treat any infection and to help with pain control.

Do I or don't I?

Over the next 6 weeks, this Manual will help you to recover and prevent problems in the future. But here are some general tips for starters:

Do

Do take exercise every day

Doctors used to think that it was dangerous to exercise if you have a heart condition. **We now know this is not true.** In fact it is important to take exercise to keep your circulation active, to give your muscles – including your heart – some exercise, and to make you feel better.

Do take a rest every day

This is important after a bypass operation. Your body does most of its repair work while you are sleeping. At the moment it has more repairs to do than usual, so you need your rest. Don't worry if you are easily tired at this stage. It's your body's way of persuading you to rest. It will pass in time.

Do pace yourself

Both overdoing it and underdoing it are bad for your recovery. Don't try to be superhuman, but don't be afraid to do things if you feel ready. Decide what you can do easily. Try it out, and if you found it pretty easy, do a bit more tomorrow. If it was too difficult, do a bit less next time. The weekly programme includes a lot of information about setting targets for yourself.

Do your relaxation

Many people who have had a procedure are anxious or have some concerns at this stage. This is quite normal. But it can make you more aware of pain or of your breathing and heartbeat. Sometimes this stress leads to people being taken back to hospital just to make sure that everything is OK. The relaxation CD/audio which comes with this Manual really does help.

Do follow the Manual

It will tell you what changes you need to make to change your lifestyle. It is important to make these changes now. As the memory of your procedure fades, your determination may fade, too.

Don't

Don't overdo things

Don't do things which make you so breathless you can't speak.

Don't lift, push or pull heavy objects that make you strain, grunt or hold your breath such as moving heavy pots, full rubbish bags, furniture or heavy shopping. This is particularly important during the first 3 months after a bypass operation. If you have to hold your breath to lift something, it's far too heavy.

Don't drive yet

Please see driving information on page 20 (Part 1 of the Manual).

Why is exercise so important?

Week 1

Exercise can:

Help...	Reduce your chances of...
 increase blood flow to your heart muscle	 type II diabetes
 reduce stress, make you feel happier and sleep better	 heart problems in the future
 lose weight, but you must change your eating habits	 falls, depression etc.
 lower your cholesterol	 joint and back pain
 reduce angina and breathlessness	 cancers (colon and breast)



Playing darts in the pub may be enjoyable and is therefore good for you, but it doesn't count as exercise!

Below are some of the activities that people with coronary artery disease have taken up. Tick any that you think might interest you. Add any others that you think you might like to try. But only **think** about them this week – we don't suggest you try them until a little later on in the programme, when you are feeling better.

- | | | |
|---------------------------------------|------------------------------------|---|
| ☐ Hill walking | ☐ Swimming | ☐ Exercise classes |
| ☐ Jogging | ☐ Badminton | ☐ |
| ☐ Bowling | ☐ Football | ☐ |
| ☐ Cycling | ☐ Dancing | ☐ |

Remember that if your exercise does not cause small increases in breathing or heart rate, it won't make you fitter. This is why you can be unfit even if you have an active lifestyle.

The Exercise/Activity Plan

On page 46 of the Manual you will find advice about how to start your walking plan.

On pages 175-179 you will find a **three** sets of exercises and some stretches.

You can also find a demonstration video of the home exercises online at:

www.heartmanual.scot.nhs.uk/index.php/download/resources

Read the instructions carefully. The Exercise/Activity Plan is suitable for most people using this Manual. However, if you have had an angioplasty and were physically active prior to your procedure, you may find the recommended starting point for the exercises too easy. If so, you may decide to set yourself a higher target. It is important to exercise regularly. Use the advice on Page 46 to set realistic targets. If there is anything you don't understand, ask your facilitator about it.

The plan is in four stages.

- | | | |
|----------------|------------------------|--|
| Stage 1 | Weeks 1 & 2 | Getting started on your plan |
| Stage 2 | Week 3 onwards | Gradually building up your plan |
| Stage 3 | Week 5 onwards | Thinking about lifelong exercise/activity |
| Stage 4 | Week 6 onwards | Regular lifelong exercise/activity |

If you had a bypass, remember that it can take up to 12 weeks for your breastbone to heal properly. So you should wait until then before taking up exercise which could affect the healing process – like golf, swimming, fishing or heavy gardening.

Important things to remember about exercise

Week 1

Follow the Exercise Plan

Trying to do too much too soon may slow down your recovery.

Use common sense

Always warm up your muscles before exercising, and again cool down afterwards. Don't exercise for at least an hour after eating a meal and avoid exercising if you are feeling unwell.

If you have been feeling unwell

If you have been feeling unwell it is best to avoid too much physical activity. When you feel better gradually return to your walking plan. Set your target at a level that is easy and build up.

You will probably have good days and bad days

On a good day, don't be tempted to do more than the plan. On a bad day, try not to put off doing your plan- for that day cut it back and aim for half of what you would usually do. Ask yourself if you've been doing a lot more than usual over the past few days. This may explain why you feel a little tired. Sometimes you may not be able to identify a trigger for a bad day.

Exercise should make you breathe harder and feel warmer

This is normal and should be your guide to how much your fitness is improving. When exercising, a general rule for everyone is **not** to exercise until you are so breathless that you can't speak – you should be able to speak while exercising. You can find out more about exercise on page 120-122 of the Manual.

If you feel chest discomfort while exercising

Stop and take a rest. Chest discomfort can sometimes be caused by setting your target too high too soon. If however you think it is angina use your GTN spray as prescribed and discuss these symptoms as soon as possible with your facilitator or doctor.

Filling in exercise records and daily activity sheets is important for two reasons

- 1** So that you can increase the amount you are doing in a safe way.
- 2** By looking at the sheets, you and your facilitator and the people you live with will be able to tell how well you are recovering.

Filling in the walking and exercise records can be very useful.

Trying to remember what you were able to do a week ago is very difficult. So write it down as you go along.

You will be asked to score your effort on the records provided. The right side of the scale refers to when you have found an activity hard (fairly hard or too hard) while the left side is when the activity has been easy (fairly easy or too easy).

Week 1

Walking record

You will have been doing some walking in hospital, so please continue as advised on discharge.

Here is a chart for recording your walks. Mark how easy you found it. If you score ‘easy’ or ‘hard’ for 2 days in a row, you should choose a new distance that is halfway between ‘fairly easy’ and ‘fairly hard.’ Go on with this target until it becomes ‘fairly easy.’ Then choose another, and so on.

Date 22nd September	I think I can easily	Walk around the room/house
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	I think I can easily	
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	I think I can easily	
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	I think I can easily	
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	I think I can easily	
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	I think I can easily	
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	I think I can easily	
That was	Too Easy Fairly Easy Fairly Hard Too Hard	

Week 1

Exercise record

Continue to exercise according to what you were advised on discharge. Each time fill in the sheet to record how hard you found the exercises, how many times you did them, and how long you were doing the exercises for. If you find the exercises ‘easy’ or ‘hard’ for 2 days in a row, increase or decrease them. You should try to do the exercises twice a day.

Date 22nd September	How long exercised 3 minutes	How many times twice
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	How long exercised	How many times
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	How long exercised	How many times
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	How long exercised	How many times
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	How long exercised	How many times
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	How long exercised	How many times
That was	Too Easy Fairly Easy Fairly Hard Too Hard	
Date	How long exercised	How many times
That was	Too Easy Fairly Easy Fairly Hard Too Hard	

Write down the things you do and how you feel each day

Date 22nd September Feel a bit low, did relaxation, feel much better
Date
Date
Date
Date
Date
Date

What is relaxation?

Deep relaxation, the kind described on the CD/audio which comes with this Manual, is the letting go of tension in body and mind.

When we are relaxed, the **adrenaline** level in our bloodstream goes down.

This makes us feel calm. It also has important benefits for the heart, especially after a heart attack or cardiac procedure.

What is bad about tension?

Muscles that are tense are working hard, even though they are not moving. This is why being tense can make you feel as exhausted as if you've just run a marathon, even though you've been sitting in a chair all day!

The muscle tension can cause unusual aches and pains, headaches, stiff neck and unusual **chest pains** – in fact, pains almost anywhere.

The unnecessary work raises the blood pressure. High blood pressure is one of the main causes of coronary artery disease.

It is important for everyone to have times when they relax during the day – not just for people who have a heart condition, but for good health in general.



Relaxation

What can relaxation do for me?

It has been proved to help people recovering from heart attack and can help after a cardiac procedure.

It is the first step in learning to control stress.

Relaxation has many benefits for health, as well as helping you to:

- ☐ reduce stress
- ☐ cope with problems
- ☐ increase your feeling of being in control
- ☐ lower your blood pressure and heart rate.

It takes time. At first it may seem that very little is happening – but stick with it!

Relaxation can also:

- ☐ reduce pain
- ☐ help you to get good refreshing sleep
- ☐ help you to cope better with work
- ☐ help to reduce tension
- ☐ help you to stop smoking if you are a smoker
- ☐ help if you have given up smoking and feel tense
- ☐ reduce fears and anxieties
- ☐ help you to keep calm in difficult situations
- ☐ help with exercise – many top sportsmen and women employ psychologists to teach them relaxation: it improves their control, their concentration and their stamina
- ☐ reduce angina.

How long does it take?

It takes most people about 12–16 weeks to get the full benefit of relaxation

To start with, you should practise for 20 minutes twice a day.

Eventually it will become quite natural – you will be able to do it whenever, or wherever, you want.

It's like learning any new skill – for example, driving, swimming, playing snooker or knitting – it takes time to get good at it. In time, you will be able to relax quickly, anywhere, without using the relaxation CD/audio.

It's a mistake to be impatient – you can't relax if you are trying to force it to happen.

People who practise get good at it.

Instructions for using the relaxation CD/audio

Make sure that no one will interrupt you. Sit or lie down somewhere comfortable.

Find somewhere quiet and warm (it may help to dim the room lighting)

If you are using a chair it must be big enough to support your whole body when you are completely floppy.

If you would not feel safe falling asleep in the chair, it is not providing enough support. The idea is to be able to let every last bit of tension out of your muscles. If you are having to carefully balance your arm on the arm of the chair, you will still be using your muscles to keep your arm there, and they won't be completely relaxed.

It may help to put a thin pillow or cushion behind your neck for your head to rest on. The main thing is that you are as comfortable as possible when relaxed.

If you are not comfortable, try relaxing somewhere else – perhaps on the sofa or the floor.

Start the relaxation CD/audio at track 1

If your mind wanders, it doesn't matter.

Don't worry or get annoyed. Just start listening again and carry on.

Remember – relaxation takes time to learn. It can't be forced.

You can't make it happen by trying hard. Trying hard is the opposite of being relaxed.

It doesn't always work. How relaxed you feel after listening to the CD/audio can depend on how you felt at the beginning.

But don't give up. Remember that it has been proven that relaxation can help people manage their heart disease.

You can find the relaxation audio online at:

www.heartmanual.scot.nhs.uk/index.php/download/resources

or download the HM Relax app via Apple or Google Play stores or scan the QR code. Please ask your facilitator for the username. The password is relax.



Ready to start?

Try playing the relaxation CD/audio now